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Example 1

A movie stuntman drives a motorcycle off a 50 m high cliff horizontally. How fast must the motorcycle leave the cliff-top to land on level ground, 90 m away from the base of the cliff where the cameras are located?

First make a sketch, and indicate your choice of coordinate system.

$$\theta_i = 0$$

$$v_{xi} = v_i \cos \theta_i = v_i$$

$$v_{yi} = v_i \sin \theta_i = 0$$

$$a_x = 0$$

$$a_y = 0$$



$$R \quad x_R = 90\text{m}; \quad y_R = -50\text{m}; \quad t_R = ?$$

Our kinematic eq's are:

(2)

$$y = \cancel{y_i} + \cancel{v_{yi}t} + \frac{1}{2}a_y t^2 = -\frac{1}{2}gt^2$$

$$x = \cancel{x_i} + \cancel{v_{xi}t} + \frac{1}{2}\cancel{a_x t^2} = v_i t$$

$$v_y = \cancel{v_{yi}} + a_y t = -gt$$

$$v_x = \cancel{v_{xi}} + \cancel{a_x t} = v_i$$

Apply the choice of coordinate system and initial value to the eq's above.

For point R; from the simplified eq.'s above

$$y_R = -\frac{1}{2}gt_R^2 \quad -①$$

$$x_R = v_i t_R \quad -②$$

$$\begin{aligned} \text{From ①: } t_R &= +\sqrt{-\frac{2y_R}{g}} \\ &= +\sqrt{-\frac{2(-50\text{m})}{+9.8\text{m/s}^2}} \end{aligned}$$

$$t_R = 3.2\text{ s}$$

(3)

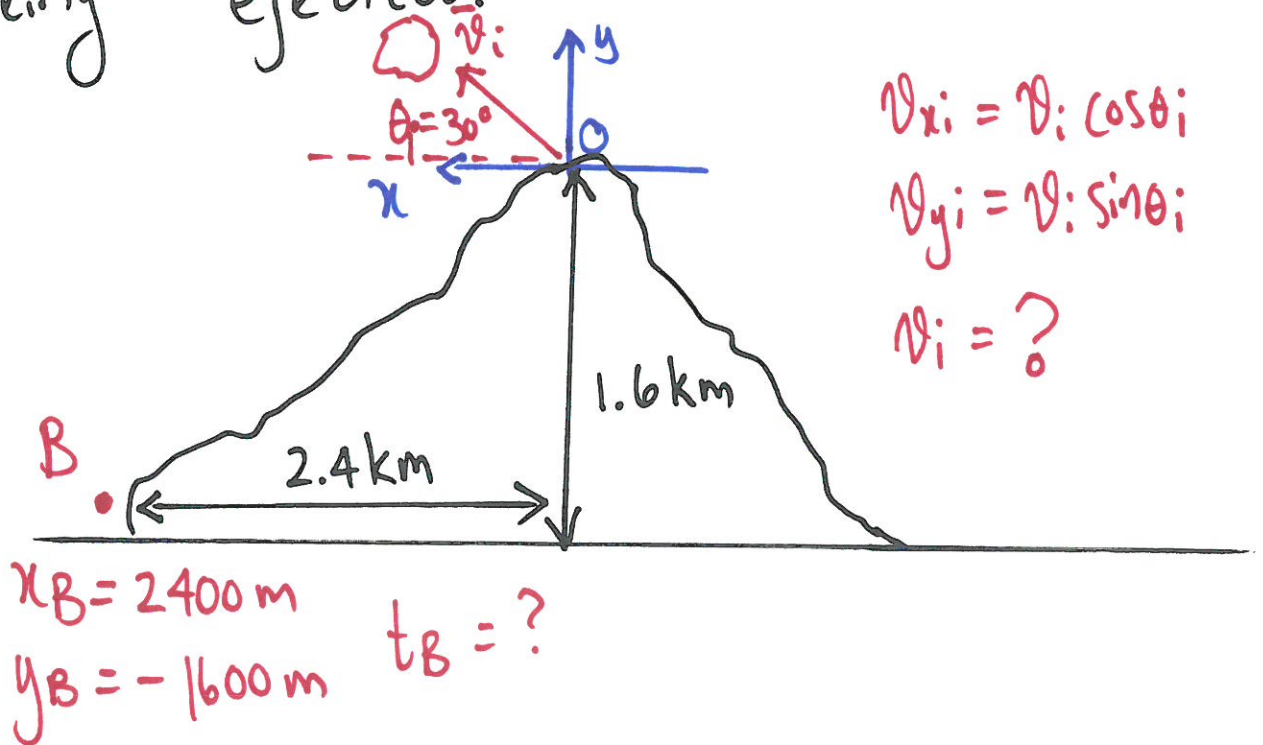
$$\begin{aligned}\text{From eq. (2): } v_i &= \frac{x_R}{t_R} \\ &= \frac{90\text{m}}{3.2\text{s}}\end{aligned}$$

$$\underline{v_i = 28.1 \text{ m/s}} \rightarrow$$

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Example 2

In the movie "Volcano", solid chunks of rock, called lava bombs, were ejected from the growing volcano. Consider a volcano, shown below, with a lava bomb being ejected.



- (a) What would the magnitude of the initial velocity v_i , at the top of the volcano, have to be in order for a lava bomb to land at the base (point B) of the volcano? Assume the lava bomb was ejected

at the angle shown in the sketch.

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$$x_B = v_{xi} t_B$$

$$x_B = v_i (\cos \theta_i) t_B$$

$$t_B = \frac{x_B}{v_i \cos \theta_i} \quad - \text{①}$$

$$y_B = v_{yi} t_B - \frac{1}{2} g t_B^2 \quad - \text{②}$$

① in ②:

$$y_B = \cancel{v_i} \sin \theta_i \times \frac{x_B}{\cancel{v_i} \cos \theta_i} - \frac{1}{2} g \frac{x_B^2}{v_i^2 \cos^2 \theta_i}$$

$$-y_B = -\tan \theta_i \times x_B + \frac{1}{2} g \cdot \frac{x_B^2}{v_i^2 \cos^2 \theta_i}$$

$$(\tan \theta_i \cdot x_B - y_B) v_i^2 = \frac{1}{2} g \cdot \frac{x_B^2}{\cos^2 \theta_i}$$

$$v_i^2 = \frac{1}{2} \cdot \frac{g x_B^2}{\cos^2 \theta_i [\tan \theta_i \cdot x_B - y_B]}$$

$$v_i^2 = \frac{1}{2} \cdot \frac{(9.8 \text{ m/s}^2)(2400 \text{ m})^2}{(\cos^2 30^\circ) [\tan 30^\circ \cdot 2400 \text{ m} - (-1600 \text{ m})]}$$

$$v_i^2 = 12604 \text{ m}^2/\text{s}^2$$

$$\underline{v_i = 112.3 \text{ m/s}}$$

check units!

(3)

(b) What would be the time of flight of this projectile from the top of the volcano to its base?

$$x_B = v_{xi} t_B$$

$$t_B = \frac{x_B}{v_{xi}} = \frac{x_B}{v_i \cos 30^\circ}$$

$$t_B = \frac{2400 \text{ m}}{(112.3 \text{ m/s})(\cos 30^\circ)}$$

$$t_B = 24.7 \text{ s} \rightarrow$$